

Collaborative Measurement-Driven Digital Twin with Data Extrapolation for DRL-assisted Optical Networks

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Abstract We present a measurement-driven digital twin framework that integrates multiple lab-based emulations with AI-based extrapolation, offering a practical approach to data-driven digital twin construction. Using 5% measured data, it achieves below 0.05 dB error and enables DRL-based spectrum allocation with 0.4 dB Q-factor improvement. ©2026 The Author(s)

Introduction

Digital twins (DTs) have emerged as an essential enabler for dynamic operation in optical networks, supporting functions such as fault management, network planning, and performance optimisation [1, 2]. Most existing DT implementations rely on simplified analytical or physics-based models, which provide scalability but lack the ability to capture system uncertainties and real operational variability [3]. Developing a fully data-driven DT, capable of reflecting these dynamics, however, requires an effective strategy to measure, sample, and collect representative data from physical networks. In commercial networks, the data is sensitive and not readily available for DT formation. Recently, the growing demand for intelligent optical network management, such as deep reinforcement learning (DRL)-based resource allocation [4, 5], further intensifies this challenge, requiring an interactive and continuously evolving DT environment. In wavelength-division-multiplexed (WDM) optical networks, due to inter-channel nonlinear effects and EDFA-induced gain tilt, the quality of transmission (QoT) of a lightpath depends on the wavelength/channel configuration of the WDM signal, i.e., channel states (CSs). Therefore, for a DT of WDM networks, a key requirement for physical-layer validation and control is to model how QoT varies with CSs across different routes [2, 6]. However, characterizing a single route would require m^n tests (m is the channel count and n is the hop count), resulting in prohibitive measurement overhead and time cost. Therefore, exhaustive measurement of all possible CSs is impractical.

To this end, we propose a collaborative DT construction framework for WDM networks that integrates sampled laboratory emulation with AI-based data extrapolation to estimate the Q-factor of all routes under all CSs. Based on a target network, its routes are classified according to their transmission characteristics and assigned to multiple testbeds for emulation, thereby improving sampling efficiency and accelerating DT construction. Each testbed uses limited optical components and purposefully sampled measurements. Subsequently, multiple neural networks (NNs) are

trained on partial measurement data to extrapolate the complete route/CS/Q database. This extrapolation module is referred to as Extra-Net. An interactive environment is then established based on this database, enabling rapid and effective training of the DRL agent. Experimental results show that, with only 5% of the complete dataset collected, the proposed DT achieves an average Q-factor estimation error of only 0.05 dB relative to real measurements. The DRL agent trained with the proposed DT improves the Q-factor of each service by approximately 0.4 dB over a heuristic approach after only 30 minutes of training. Compared with online DRL training through direct testbed interaction, it significantly reduces training time while maintaining performance.

Collaborative Measurement-Driven Digital Twin for WDM networks

The proposed DT framework is illustrated in Fig. 1. The target network is first emulated through a collaborative platform (*step 1*). Specifically, the routes in the target network are grouped into multiple test classes according to selected criteria (e.g., hop count and transmission distance), and each class is then assigned to a corresponding testbed for measurement collection. To enable collaborative use of heterogeneous testbeds, a cross-platform standardisation step should be applied to align measurement conditions and data representations, so that differences in device types and operating settings do not hinder consistent data aggregation. The detailed emulation procedure using the testbed is shown in Fig. 2. Within each testbed, the experimental setup can be further adjusted for individual routes to reproduce their corresponding transmission characteristics, including the number of spans, fibre lengths, EDFA gain, and the configuration of reconfigurable optical add-drop multiplexers (ROADMs). For each span along a route, its transmission characteristics are emulated using a fibre segment of similar length but lower end-to-end loss, together with a variable optical attenuator (VOA) placed at the end of the fibre segment, before the pre-amplifier. The VOA is used to compensate for the loss difference between the fibre

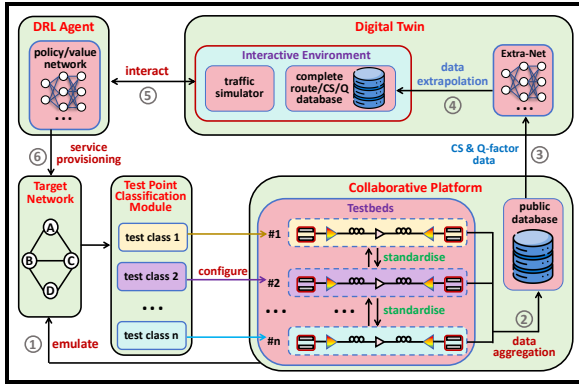


Fig. 1: Measurement-driven digital twin framework

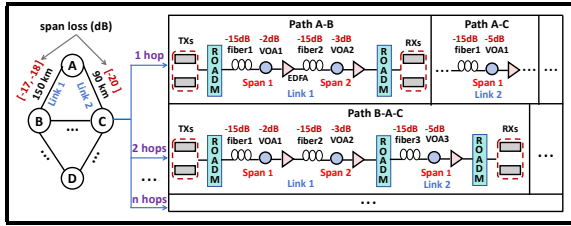


Fig. 2: Target network emulation using laboratory resources

segment and the target span. This ensures that the optical power and optical signal-to-noise ratio (OSNR) at the end of each span, each link, and at the receiver in the testbed closely resemble those in the actual system. In this study, we consider a small-scale target network in which the routes are relatively short. Under this condition, the nonlinear noise in the real and emulated routes is comparable [7]. Therefore, with proper VOA adjustment, the emulated fibre can achieve a generalised SNR (GSNR), bit error rate and Q-factor similar to those of a real link. For multi-hop routes, ROADMs are inserted between links for wavelength add/drop, to emulate intermediate nodes. Routes with the same hop count use the same ROADM configuration. By dynamically reconfiguring the transmitters and ROADMs, Q-factors under different channel combinations (i.e., CSs) can be measured. This reconfigurable setup enables the emulation of an entire target network using only h fibre segments and VOAs, where h denotes the maximum number of spans among all routes.

The data collected on each testbed is aggregated and recorded into a public database (step 2). These data are then used to train Extra-Net to capture the underlying relationship between the CSs and the Q-factor (step 3). One dedicated NN is assigned to each group of routes with the same hop count, such that a total of h NNs are used in Extra-Net. The input features of each NN include the CS of every link along the route (1 for active and 0 for inactive), along with the measured end-to-end loss of each link. Subsequently, Extra-Net is used to extrapolate the complete route/CS/Q-factor database by feeding in all possible CSs (step 4). Next, an interactive environment is established based on the route/CS/Q-factor database to train

the DRL agent. Specifically, a traffic simulator is employed to randomly generate dynamic service requests $R_t(s, d)$ between nodes s and d at each timestep t . Given the current network state s_t , the DRL agent selects an action a_t , corresponding to choosing a channel on one of the candidate routes between s and d to provision the request. After executing the action, the CSs in the network are updated, and the database is queried to obtain the average Q-factor Q_t of all active lightpaths, which serves as the agent's reward r_t . In the DRL agent design, the state s_t consists of $e \times m + k \times h$ features, where e , m , k , and h denote the numbers of links, channels, candidate routes, and max hop count among all routes, respectively. The first $e \times m$ binary elements represent the CSs of the entire network, while the $k \times h$ elements encode the current request R_t , as illustrated in [5]. The reward r_t is assigned as Q_t if the request is provisioned, otherwise -1 is given. Through continuous interaction with the environment, the DRL agent learns to maximise the cumulative reward, thereby improving the service Q-factor and reducing the BP of the network (step 5). Finally, the trained DRL agent can be deployed as a network controller for real-time service provisioning in the target network (step 6).

Experimental Setup and Results

We emulate a single-span 4-node target network with the topology shown on the left side of Fig. 2. We consider $k = 2$ candidate routes with a maximum hop count of $h = 2$, and assume that the network consists of three types of links with losses 0 dB, 2 dB, and 3 dB higher than those of the testbed fibres, respectively. Two 75 km laboratory fibre segments and VOAs are used to emulate all routes in the network. The testbed setup for collecting data over each 2-hop route is illustrated in Fig. 3. Eight dense WDM (DWDM) optical channels carrying 32 Gbaud real-time Nyquist PM-16QAM signals with a 50 GHz channel spacing, ranging from 193.75 THz to 194.10 THz, are generated using two Facebook Voyager coherent transponders and multiplexed by a 4×16 wavelength-selective switch (WSS). The 8 DWDM channels can be added at either the initial or intermediate WSSs, representing different spectrum allocation in the first and second fibre segments, respectively. The DWDM signal is demultiplexed at the terminal, and the BER of each channel is measured by the coherent receivers through an HTTP-based REST API, from which the corresponding Q-factor is calculated. By adjusting the WSS configurations, the wavelength components of the WDM signal in each fibre segment, and thus the CSs on each link, can be dynamically reconfigured. For a single-hop route, only one pair of WSSs is required at the transmitter and receiver

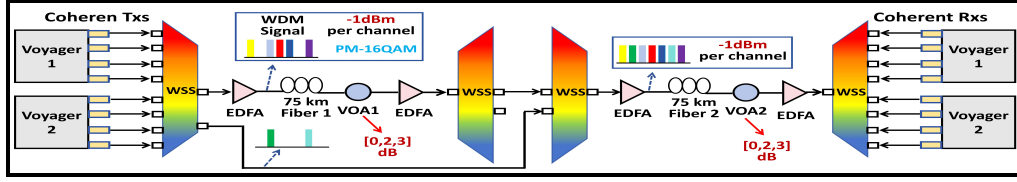


Fig. 3: Experimental setup for channel state-Q-factor measurement on the testbed.

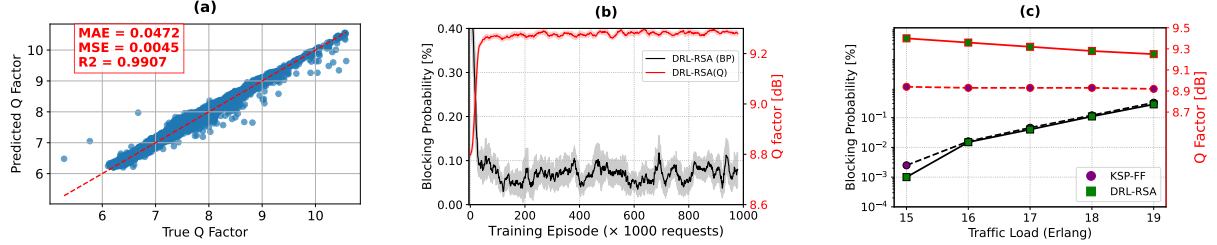


Fig. 4: Validation of the proposed DT framework: (a) Q-factor predicted by Extra-Net vs. measured values; (b) performance of DRL agent during training; (c) performance of DRL and heuristic under varying traffic loads on testbed.

Tab. 1: MAE (dB) and R^2 vs. training data ratio

Training ratio (%)	1	5	20	40
1-hop (MAE / R^2)	2.56 / -14.25	0.39 / 0.47	0.07 / 0.96	0.04 / 0.99
2-hops (MAE / R^2)	0.08 / 0.97	0.05 / 0.99	0.04 / 0.99	0.05 / 0.99

nodes to configure the CSs on the link. The launch power of each channel is equalised to -1 dBm, and the gain of each pre-EDFA is kept identical across all scenarios to ensure consistent transmission conditions.

For the target network considered in this study, Extra-Net consists of two NNs trained on the measured data to extrapolate the Q-factors for the one-hop and two-hop scenarios, respectively. Each NN has 2 hidden layers with 32 neurons per layer. Table 1 summarises the mean absolute error (MAE) and coefficient of determination (R^2) achieved by the NNs with different training ratios, where the training ratio denotes the proportion of data used for training relative to all possible CSs. Because the total number of CSs (2^8) in the one hop case is small, a low training ratio corresponds to a limited amount of data, resulting in underfitting of the NN. However, as the amount of training data increases, the network performance improves significantly. In contrast, for the two-hop case with a higher number of CSs, the NN can still achieve a mean MAE below 0.1 dB and an R^2 value above 0.95 even with a low training ratio (e.g., 1%). Extra-Net, with NNs trained on 20% of the data for 1-hop routes and 5% of the data for 2-hop routes, is evaluated on a validation set containing all unseen data measured on the testbed. As shown in Fig. 4(a), the average Q-factor prediction error is below 0.05 dB and R^2 reaches 0.99. This indicates the strong generalisation capability of Extra-Net in a narrow-band DWDM system, enabling up to 95% reduction in data measurements. With domain adaptation techniques [8], the DT can potentially scale to full C-band systems.

Extra-Net is then used to construct the route/CS/Q-factor database for training the DRL agent. The DRL agent is trained using the hyperparameters reported in [5]. Figure 4(b) shows the

performance of the DRL agent during training. As training progresses, the agent is able to simultaneously reduce the BP and improve the Q-factor until convergence. The training process is completed in approximately 30 minutes. The performance of the trained DRL agent is evaluated and compared with that of the k-shortest-path first-fit (KSP-FF) algorithm on the testbed. Specifically, the CSs of all routes at each timestep are recorded during the routing and spectrum allocation processes of the two algorithms in the simulation environment. These recorded CSs are then reproduced on the testbed, and the corresponding average Q-factors of all lightpaths are measured to assess the performance that the two algorithms can achieve in the target network. As shown in Fig. 4(c), the DRL agent achieves an approximately 0.4 dB improvement in Q-factor compared to KSP-FF across all evaluated traffic loads. Meanwhile, it also exhibits slightly better performance in terms of BP. These results indicate that the DT can rapidly train an effective DRL agent for real optical networks. This method effectively addresses the challenge of high training complexity in conventional online training caused by large data requirements and slow measurement acquisition.

Conclusions

We propose a DT framework for WDM networks, in which the target network can be emulated collaboratively across globally distributed testbeds using laboratory resources. Partial data are collected through real measurements on the experimental platforms and then combined with AI-based data extrapolation to construct a complete target-network dataset. The resulting DT enables the rapid training of an effective DRL agent for use in the real target network. Due to the constraints of available laboratory resources, the target network emulated in this work is relatively small. As future work, we will validate the scalability of the framework in larger networks with more channels.

Acknowledgements

The authors acknowledge the support of EU-funded project ECO-eNET (No. 10113933.) and the UK EPSRC project HASC (No. EP/X040569/1).

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